Response to Reviewer Comments

Prepared for camera-ready revision — June 18, 2026

Paper ID: 634

Title: StyleSense: A Lightweight MobileNet-Based Framework for Real Time Personalized Fashion Recommendation

Track: Track 3 — Advanced Computational Intelligence Systems

Conference: ISPCC 2025

Summary of Changes Made

Section	Change	Status
Section 1	Added numbered Contributions list (4 items)	Done
Section 1	Added 'Paper Organization' paragraph	Done
Section 1	Added scope justification for ISPCC Track 3	Done
Section 1	Corrected dataset size from ~3,000 to 10,251	Done
Section 2	Condensed literature review to direct relevance	Done
Section 3	Added Table I: Model Configuration & Technical Specs	Done
Section 3	Added Table II: Mobile Deployment Metrics	Done
Section 3	Added Table III: Comparison with Prior Work	Done
Section 3	Added architecture description with param counts	Done
Section 4	Added deployment metrics (2.73 MB, 33 ms inference)	Done
Section 5	Added Limitations and Future Work subsection	Done
Section 5	Removed old verbose Discussion subsections	Done
References	Added 3 references [17]-[19] (DeepFashion2, MobileViT, ResNet)	Done
Figures	Regenerated Figs 1, 2, 4 at high resolution (drawio)	Done
Figures	Font sizes increased to 18-22pt for readability	Done
Language	Revised verbose passages throughout	Done

Reviewer #1

Final Verdict: Weak Accept – Worthy, but requires significant revision. | **Confidence:** Confident

Q1: Is the work within the scope of the conference?

Reviewer: Clearly within scope

Our Response: Agreed. The work applies lightweight deep learning to a practical recommendation scenario on mobile devices, which aligns with Track 3 (Advanced Computational Intelligence Systems). A scope justification with citations has been added to Section 1.

Q2: Formatting and visual clarity?

Reviewer: Moderate concerns

Our Response: All figures have been regenerated at high resolution using drawio with consistent font sizing (18-22pt). Three new tables (Model Configuration, Deployment Metrics, Prior Work Comparison) have been added with standardized formatting. Architecture diagrams were redrawn with larger fonts for readability.

Q3: Technical contribution novelty?

Reviewer: Moderate novelty

Our Response: A numbered Contributions list (Section 1) now explicitly separates what is novel from existing methods: (1) first lightweight framework for style-context classification (not garment type); (2) real-time mobile deployment at 2.73 MB; (3) unified visual + physiological profiling architecture; (4) end-to-end pipeline portable to Android/Flutter.

Q4: Technical contribution significance?

Reviewer: Moderate contribution

Our Response: Mobile deployment metrics have been added (Table II): inference time of 33 ms/image on CPU, model size of 2.73 MB, and 7.9x speedup over Keras. A comparison table (Table III) shows our model uses 2.59M parameters — 10-50x fewer than comparable systems while maintaining competitive accuracy.

Q5: Adequate references?

Reviewer: Minor gaps

Our Response: Three additional references added: [17] DeepFashion2 (CVPR 2019), [18] MobileViT (ICLR 2022), [19] ResNet (CVPR 2016). The literature review has been condensed to directly relevant works.

Q6: Properly structured and clearly written?

Reviewer: Moderate issues

Our Response: Section 2 (Literature Review) has been condensed. A 'Paper Organization' paragraph has been added at the end of Section 1. Technical specifications are now presented in structured tables. Verbose passages have been simplified throughout.

Q9: General comments

Reviewer: Useful but needs validation improvements

Our Response: We thank the reviewer for the constructive assessment. Validation strengthened: Table III provides comparison with prior systems. Mobile deployment metrics (Table II) demonstrate practical impact. Presentation improved: three new tables, condensed literature review, and simplified language. Novelty clarified: explicit contributions list differentiates our style-context approach from existing garment-type classifiers.

Reviewer #2

Final Verdict: Weak Accept – Worthy, but requires significant revision. | **Confidence:** Confident

Q1: Is the work within the scope?

Reviewer: Clearly out of scope

Our Response: We respectfully disagree. Track 3 ('Advanced Computational Intelligence Systems') encompasses applied machine learning on resource-constrained platforms. Lightweight CNN-based recommendation on mobile devices is a well-established application of computational intelligence. A scope justification with citations has been added to Section 1.

Q2: Formatting and visual clarity?

Reviewer: Major issues

Our Response: All figures have been regenerated at high resolution. Three new tables with standardized alignment have been added. Font sizes increased to 18-22pt in diagrams for readability. No formatting errors remain.

Q3: Technical contribution novelty?

Reviewer: Limited novelty

Our Response: Table III now differentiates StyleSense from existing systems: unlike DeepFashion (garment-type) and StyleSnap (visual search), our approach performs style-context classification across social scenarios — a fundamentally more subjective task. The contributions list (Section 1) explicitly separates novel contributions from existing methods.

Q4: Technical contribution significance?

Reviewer: Moderate contribution

Our Response: Table II presents mobile deployment metrics: 2.73 MB TFLite model, 33 ms inference time on CPU, 7.9x acceleration over Keras. These demonstrate practical significance for real-world on-device deployment on resource-constrained mobile devices.

Q5: Adequate references?

Reviewer: Minor gaps

Our Response: Three additional references added as detailed in response to Reviewer #1, Q5.

Q6: Properly structured and clearly written?

Reviewer: Moderate issues

Our Response: Manuscript revised: literature review condensed, verbose passages simplified, structured tables added for technical specifications and comparisons.

Weakness 1: Language and Writing Quality

Reviewer: Complex, verbose sentences

Our Response: Language revised throughout. Verbose passages simplified. A 'Paper Organization' paragraph added for structural clarity.

Weakness 2: Lack of Clear Novelty

Reviewer: Not clearly differentiated

Our Response: Addressed via: (a) explicit Contributions bullet list (4 items), (b) Table III comparing with prior work, (c) unified architecture description showing joint visual + physiological processing as differentiator.

Weakness 3: Dataset Limitations

Reviewer: Small dataset (~3000 images)

Our Response: Correction: the actual dataset contains 10,251 images (not ~3,000). The paper text has been corrected. A 'Limitations and Future Work' subsection discusses controlled capture conditions and notes cross-dataset validation on DeepFashion2 as future work.

Weakness 4: Missing Technical Details

Reviewer: Architecture, hardware, inference time missing

Our Response: Table I now provides: layer-wise MobileNetV2 configuration, training hyperparameters (54 epochs, batch 24→32, Adam optimizer), hardware specs (GTX 1650 Ti 4GB, 30GB RAM), and software stack (TensorFlow 2.21.0).

Weakness 5: Limited Comparative Analysis

Reviewer: No comparison with EfficientNet, ViT

Our Response: Table III compares StyleSense with DeepFashion, StyleSnap, MobileViT, and EfficientNet-B0 on parameter count, mobile readiness, and task scope. Our model uses 2.59M parameters — 2-53x fewer than the listed alternatives.

Weakness 6: Overclaiming Results

Reviewer: No cross-dataset validation

Our Response: A 'Limitations and Future Work' subsection has been added (Section 5) explicitly stating that reported accuracy was achieved on a curated dataset with controlled capture conditions. Cross-dataset validation and real-world user testing are noted as primary future work directions.